

Tanishk Yadav

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EDUCATION

New York University, Tandon School of Engineering

Brooklyn, NY

Master of Science in Financial Engineering

05/27

- Coursework: Introduction to Derivative Securities, Valuation, Quantitative Methods, Credit Risk and Financial Risk Management, Econometrics and Time Series Analysis, Real Time Risk Management, Algorithmic Trading and High Frequency Finance, Corporate Valuation

SRM University (100% scholarship)

Amaravati, India

B.Tech in Computer Science & Engineering (AI & Machine Learning)

06/25

- Coursework: Single/Multi Variable Calculus, Differential Equations, Linear Algebra, Probability, Statistics, Economics

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, R, Java, JavaScript
- **Libraries & Frameworks:** Pandas, NumPy, SciPy, Statsmodels, Scikit-Learn, TensorFlow/Keras, XGBoost, Matplotlib
- **Financial Concepts:** Options Pricing (Black-Scholes, Greeks), Monte Carlo Simulation, VaR/CVaR, Regime Switching (HMM), Tail Risk Hedging, Bootstrap Resampling, Time Series (ARIMA, AR), Mean-Variance Optimization
- **Tools & Platforms:** Bloomberg Terminal, Git, LaTeX, Jupyter, Excel, VS Code
- **Certifications:** Bloomberg Market Concepts (BMC), Bloomberg Finance Fundamentals, Akuna Capital Options 101, Financial Markets (HONORS), AI for Investments, IBM Machine Learning

EXPERIENCE

IUDX (India Urban Data Exchange), Bengaluru, India

06/23 – 08/23

Summer Intern, Framework Development

- Developed a secure, cross-platform digital consent collector using SHA-256 cryptographic hashing, ensuring regulatory compliance for critical data aggregation on the ADeX platform—a government-academia joint venture
- Engineered full-stack JavaScript solutions for consent collection and retrieval as part of a 4-person team, ensuring 100% compliance with central govt. regulations and enabling creation of 38 critical datasets for the government

PROJECTS

Regime-Aware Fundamental Strategy & Point-in-Time Data Engine | *Python, HMM, SEC EDGAR*

- Architected a production-grade SEC EDGAR data pipeline enforcing zero lookahead bias at 4 independent layers (FilingsIndex, FilingSelector, XBRLParser, BloombergMapper), each gated by cutoff-date assertions; outputs normalized financials with derived metrics (ROIC, FCF yield, leverage ratios)
- Built a walk-forward HMM equity strategy across 18 years (2008–2026) delivering **633% total return net of a 2/25 fee structure**, with **8.33% annualized Fama–French 5-factor alpha** ($t = 2.67$, $p = 0.008$); defensive market beta of 0.77
- Implemented composite ranking engine using 4 fundamental factors (ROIC, leverage, earnings stability, FCF yield), each smoothed with 4-period EWMA ($\alpha \approx 0.4$); applied -30% forced-liquidation penalty after 20 consecutive NaN-price days to penalize survivorship bias from delistings
- Reduced maximum drawdown by 2.45 pp vs. SPY (-49.03% vs. -51.48%) through 4-state regime-conditional position sizing; first-half sub-period alpha of 10.34% ($p = 0.019$) confirms signal strength during crisis recovery regimes

Dynamic Regime-Based Sector Allocation & Tail Risk Hedging | *Python, Scikit-Learn, Monte Carlo*

- Built a Regime-Switching Risk Model using Hidden Markov Models (HMM) to classify market states into 4 regimes (Crisis, Bear, Transition, Bull), leveraging a custom Synthetic History Generator to backfill sector ETF data to 2005 for capturing GFC-era volatility dynamics
- Engineered a strict Walk-Forward (Out-of-Sample) backtest (2023–Present) using Ledoit-Wolf shrinkage for covariance estimation; achieved **Sharpe Ratio of 1.49** while reducing Maximum Drawdown by **46%** (-10.12% vs. -18.76% benchmark)
- Validated robustness via Parametric Bootstrapping (1,000 simulations) and Null Hypothesis Falsification, proving true alpha generation ($+\$26K$ excess return per $\$100K$ deployed) with $p < 0.05$ rejection of noise-driven outperformance

S&P 500 Sector-Level Clustering & Market Structure Analysis | *Python, Scikit-Learn, Keras*

- Analyzed 66 S&P 500 sub-sectors across 425+ companies using 5 normalized financial metrics (YoY market cap growth, revenue growth, weighted-simple variance, 6-month and 4-year sector beta) to classify market structures as monopolistic, duopolistic, or oligopolistic
- Developed a custom overperformance index using squared-weight allocation; applied Hierarchical + K-Means clustering to uncover inter-sector similarity patterns and trained an MLP neural network achieving 93% classification accuracy on held-out test data